

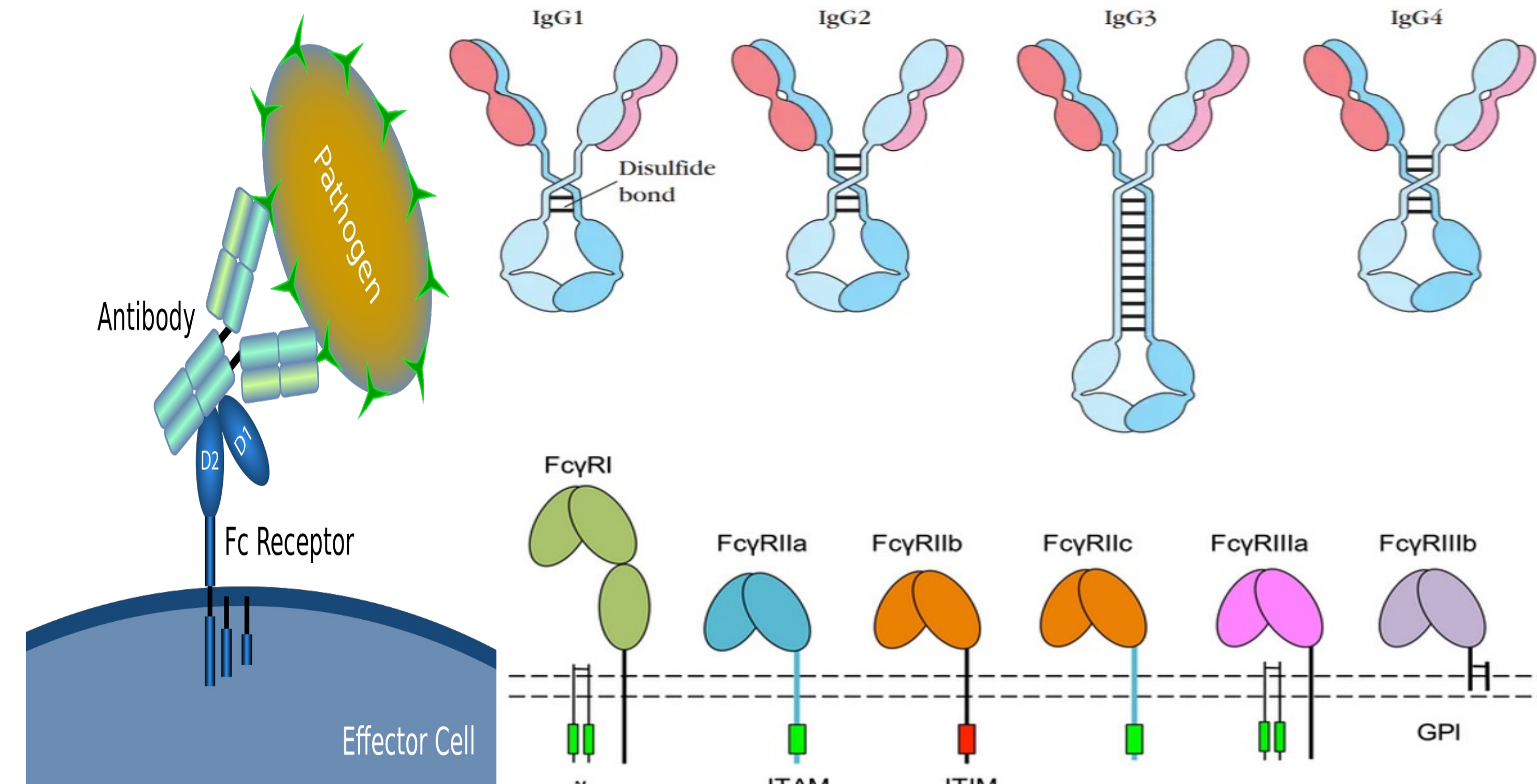
# Quantitative analysis of associations between antibody constant region (Fc) and FcR genetic variation and vaccine-elicited functional antibody responses

Song Young Oh<sup>1</sup>, Cliburn Chan<sup>2</sup>, Justin Pollara<sup>3</sup>

<sup>1</sup> Master's in Interdisciplinary Data Science, Duke University; <sup>2</sup> Department of Biostatistics and Bioinformatics, Duke University; <sup>3</sup> Department of Surgery, Duke University School of Medicine

## Background

- The HVTN108 trial evaluated the immunogenicity of a DNA prime, adjuvanted protein boost (DNA-HIV-PT123), vaccine regimen.
- We hypothesize that certain IgG and Fc receptor (FcR) alleles could affect the functional antibody response. We tested this hypothesis using sequencing data from the placebo and vaccine arms of HVTN108 together with results from BAMA, ADCC and ADCP assays.



Source: Wikipedia, <https://microbeonline.com/igg-antibody-structure-subclasses-functions-and-clinical-significance/>

## Data

- Data from the HVTN108 trial included immune responses at two different timepoints (baseline and peak response).
- Total of 334 participants in treatment (n=310) and placebo (n=24) groups enrolled in US and South Africa from June 2016 to July 2018.
- Antibody and FcR genotypes were determined by targeted PCR amplification and Illumina next-generation sequencing.
- The immune response was measured by three different assays.
  - Binding antibody multiplex assay (BAMA)
  - Antibody-dependent cellular cytotoxicity (ADCC)
  - Antibody-dependent cellular phagocytosis (ADCP)

## Method

- Exploratory data analysis examines general distributions of the genetic data through data visualization tools including bar plots, pie charts, correlation matrices, and cluster maps.
- We use linear regressions to compare the immune response levels among carriers or homozygotes of different IgG alleles and FcR alleles with and without interaction terms.
- Each p-value was adjusted by the Benjamin-Hochberg false discovery rate (FDR) correction after setting a threshold of 0.2 in the analysis.

## Results

### 1) What is the distribution of genotypes for each IgG and FcR allele?

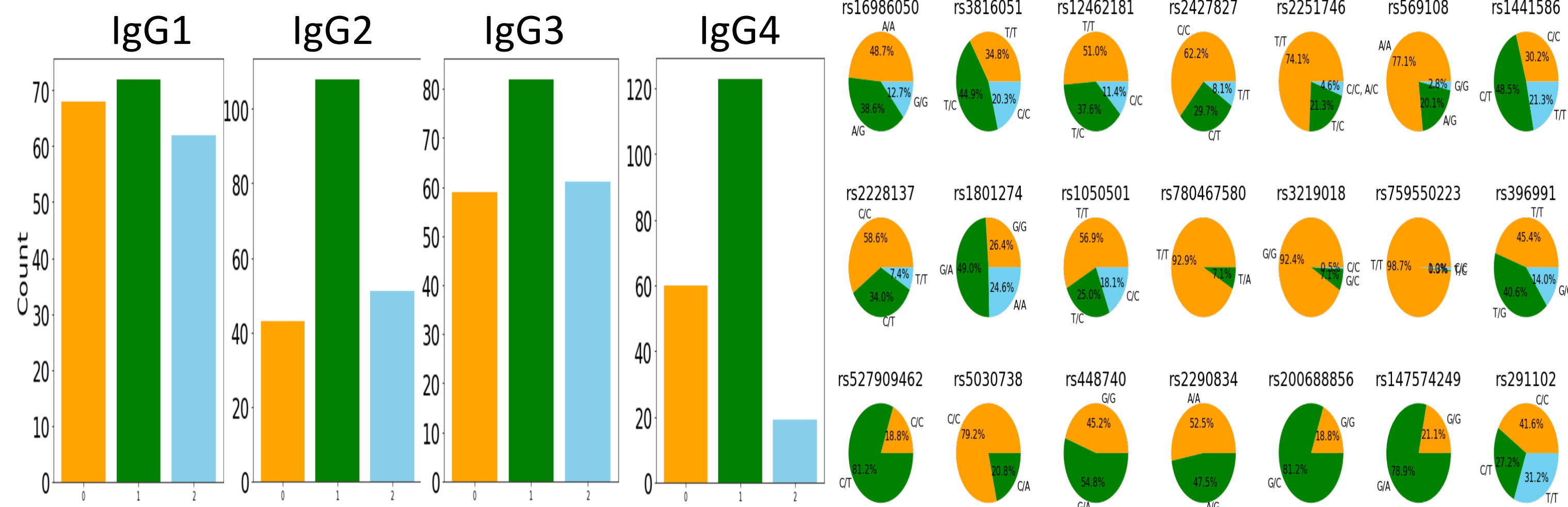


Figure 1. Frequency of wild type, carrier and homozygote in IgG1-4 alleles

Figure 2. Genotype distribution of FcR alleles

### 2) Is there any relationship between IgG and FcR alleles?

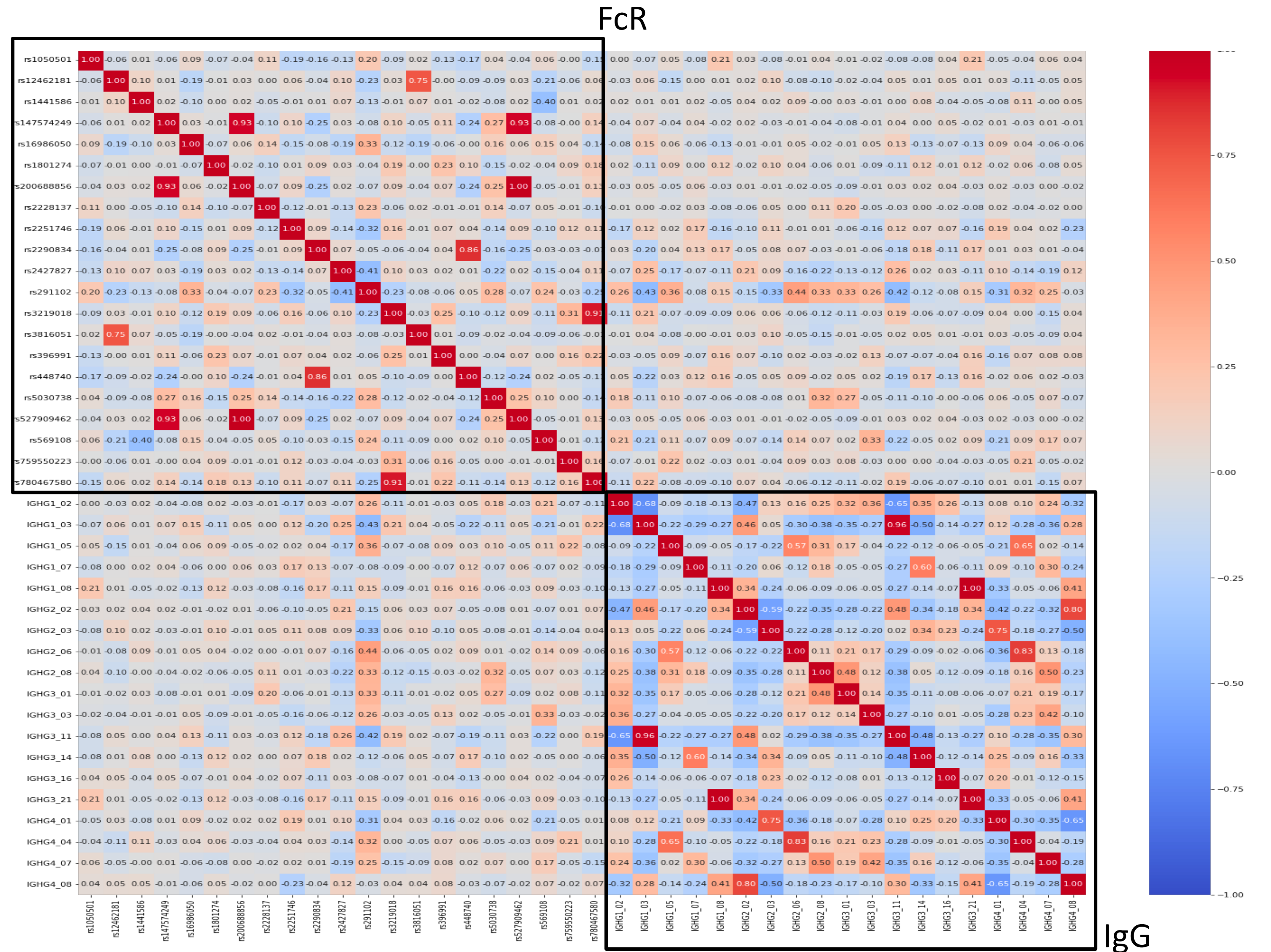


Figure 3. Correlation matrix of IgG/FcR alleles

### 3) Is there any relationship between geographic locations and antibody responses?

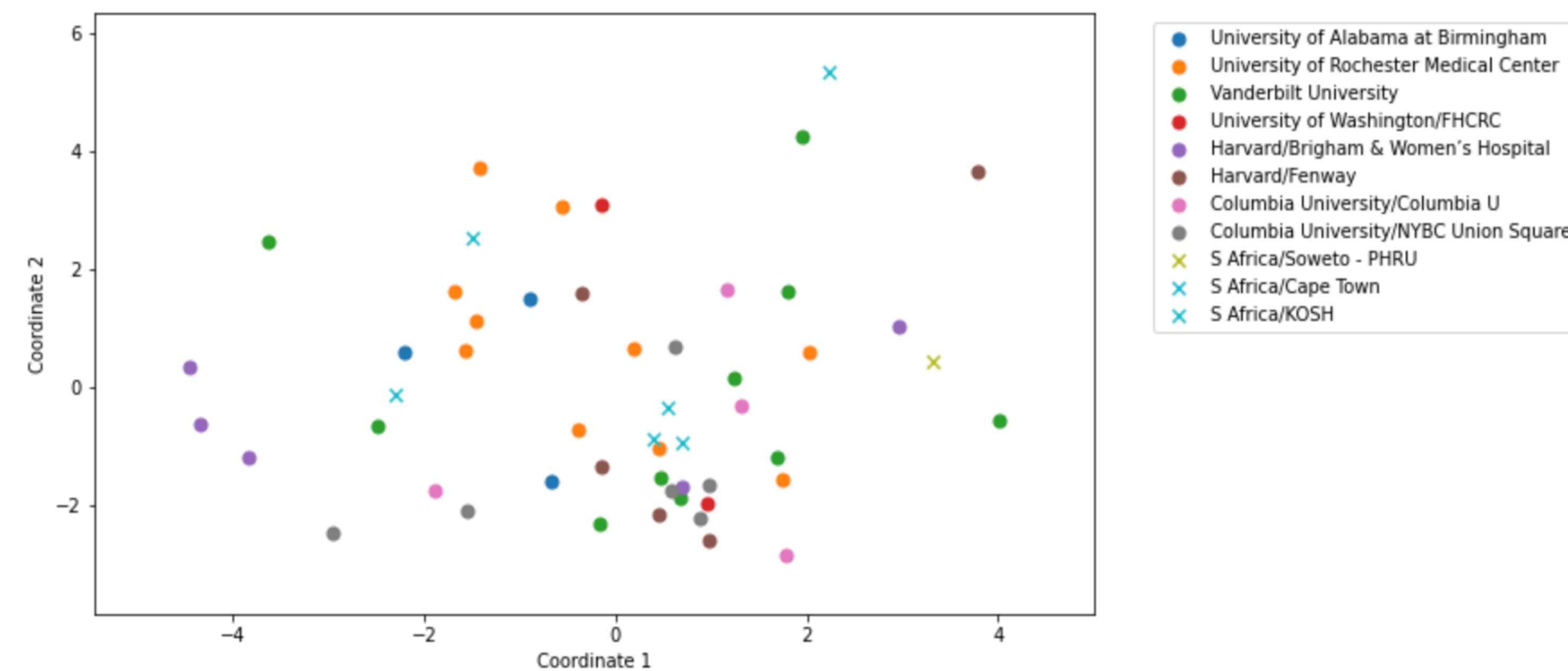


Figure 4. Multidimensional scaling (MDS) plot of antibody response (ADCC)

### 4) Do IgG/FcR alleles associate with functional antibody responses?

Table 1. Regression results of significant IgG alleles (treatment group)

| Allele   | Coefficient | FDR corrected P-Value | Assay               |         |
|----------|-------------|-----------------------|---------------------|---------|
| IGHG1_03 | -1.811872   | 0.015617              | ADCC(Titer)         | TV1 GTL |
|          | -3.097215   | 0.064792              | ADCC(Activity Peak) |         |
| IGHG3_11 | -1.796755   | 0.015617              | ADCC(Titer)         | TV1 GTL |
|          | -3.170945   | 0.064792              | ADCC(Activity Peak) |         |
| IGHG4_07 | 3.846424    | 0.015617              | ADCC(Titer)         | TV1 GTL |
|          | 6.011322    | 0.064792              | ADCC(Activity Peak) |         |
| IGHG1_07 | 2.843441    | 0.016899              | ADCC(Titer)         | TV1 GTL |
|          | 4.907869    | 0.064792              | ADCC(Activity Peak) |         |
| IGHG2_02 | -2.021771   | 0.016899              | ADCC(Titer)         | TV1 GTL |
|          | -3.218367   | 0.117306              | ADCC(Activity Peak) |         |

Table 2. Regression results of significant FcR alleles (treatment group)

| Allele             | Coefficient | FDR corrected P-Value | Assay                          |           |
|--------------------|-------------|-----------------------|--------------------------------|-----------|
| FCGR2A (rs1801274) | 0.830809    | 0.105050              | ADCC(Titer)                    | 1086C LUC |
|                    | 4.457721    | 0.109100              | ADCC(Activity Peak)            | TV1 LUC   |
|                    | 0.640371    | 0.069685              | BAMA(00MSA 4076 gp140)         |           |
|                    | 0.631427    | 0.091238              | BAMA(A1.con.env03 140 CF)      |           |
|                    | 0.630406    | 0.045801              | BAMA(Con 6 gp120/B)            |           |
|                    | 1.022237    | 0.028901              | BAMA(gp70-TV1.GSKvacV1V2/293F) |           |
|                    | 0.814255    | 0.175241              | BAMA(gp70_B.CaseA_V1_V2)       |           |

## Conclusions

- Our study explores the distribution of genotypes and allotypes for IgG and FcR genes in the specific regions and populations where HIV vaccines are being evaluated.
- The preliminary results align with our hypothesis that certain mutations in the constant region of antibodies and in the Fc receptor are meaningfully associated with vaccine-elicited immune responses, which provides insight into the genetic determinants of vaccine-elicited antibody levels and functionality.
- Particularly, our analysis indicates that the FCGR2A (rs1801274) mutation is associated with increases in ADCC, as well as higher binding antibody responses to multiple antigens.
- A comprehensive understanding of associations between human genetic variation and vaccine-elicited immune responses will help to inform the design and interpretation of future HIV-1 clinical trials.

## Acknowledgments

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